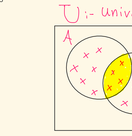
④ Intersection :- $A \cap B$ ⇒ elements that are in both A and B.

Set builder notation :- $A \cap B = \{x \mid x \in A \wedge x \in B\}$

Ex 1:- $A = \{1, 2, 3, a, 3\}$,

$B = \{2, 4, a, c\}$, then

$A \cap B = \{2, a\}$



$A \cap B$

⑤ Union :- $A \cup B$ ⇒ elements in A or in B

Set builder notation :- $A \cup B = \{x \mid x \in A \vee x \in B\}$

Ex :- based on the example 1

$A \cup B = \{1, 2, 3, a, 4, c\}$

$A \cup B$

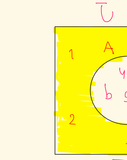
⑥ Complement :- $\overline{A}$ ⇒ all elements that are not in A.

Set builder notation :- $\overline{A} = \{x \mid x \in U \wedge x \notin A\}$

Ex 2 :- $U = \{1, 2, 3, 4, 5, a, c\}$

$A = \{4, 5, b, c\}$

$\overline{A} = \{1, 2, 3, a\}$



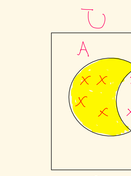
$\overline{A}$

⑦ Subtraction :- $A - B$ or $A \setminus B$ ⇒ elements in A but not in B.

Set builder notation :- $A - B = \{x \mid x \in A \wedge x \notin B\}$

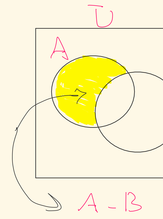
Ex :- based on example 1

$A - B = \{1, 3\}$

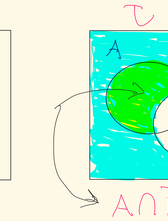


$A - B$ or $A \setminus B$

⑧ Note $A - B = A \cap \overline{B}$



$A - B$



$A \cap \overline{B}$

⑨ Set Identities :-

1. $A \cap B = B \cap A$ } commutative
   $A \cup B = B \cup A$ }
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ } distributive
   $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ }
3. $\overline{A \cap B} = \overline{A} \cup \overline{B}$ } DeMorgan
   $\overline{A \cup B} = \overline{A} \cap \overline{B}$ }
4. $A \cap \overline{A} = \emptyset = \{\}$ empty set
   $A \cup \overline{A} = U$ universe
5. $\overline{\overline{A}} = A$ complement
6. $A \cap \emptyset = \emptyset$
   $A \cup \emptyset = A$
7. $A \cup (A \cap B) = A$ } absorption
   $A \cap (A \cup B) = A$ }
8. $(A \cup B) \cup C = A \cup (B \cup C)$ } associative
   $(A \cap B) \cap C = A \cap (B \cap C)$ }
9. $A \cup A = A$ } idempotent
   $A \cap A = A$ }

⑩ Example :- prove that $A - (A - B) = A \cap B$ using set identities

note :- ⑧ ⇒ $A - B = A \cap \overline{B}$

Ans :-

$A - (A - B) = A - (A \cap \overline{B})$ by ⑧

$= A \cap (\overline{A \cap \overline{B}})$ by ⑧

$= A \cap (\overline{A} \cup \overline{\overline{B}})$ by Demorgan

$= A \cap (\overline{A} \cup B)$ by complement

$= (A \cap \overline{A}) \cup (A \cap B)$ by distribution

$= \emptyset \cup (A \cap B)$ by $A \cap \overline{A} = \emptyset$

$= A \cap B$ by $A \cup \emptyset = A$

⑪ Set builder notation :- definitions of set operations, etc.

Ex :- $A \cap B = \{x \mid x \in A \wedge x \in B\}$

note :- $\in$ ⇒ belongs to.

$\notin$ ⇒ does not belongs to.

⑫ Example :- prove that $\overline{A \cup B} = \overline{A} \cap \overline{B}$ by using set builder notation. Note X is a variable :-

Ans :- show that $x \in \overline{A \cup B} \Rightarrow x \in \overline{A} \cap \overline{B}$

① Take an arbitrary element x in $\overline{A \cup B}$, then :-

$x \in \overline{A \cup B} \Rightarrow x \notin A \cup B$ definition of complement

$\Rightarrow \sim (x \in A \cup B)$ definition of $\notin$

$\Rightarrow \sim (x \in A \vee x \in B)$ definition of union

$\Rightarrow \sim x \in A \wedge \sim x \in B$ Demorgan law (Logic) ($\sim(p \vee q) \equiv \sim p \wedge \sim q$)

$\Rightarrow x \notin A \wedge x \notin B$ definition of $\notin$

$\Rightarrow x \in \overline{A} \wedge x \in \overline{B}$ definition of complement

$\Rightarrow x \in \overline{A} \cap \overline{B}$ definition of intersection

$\therefore \overline{A \cup B} \Rightarrow \overline{A} \cap \overline{B}$ ✓✓

⑬ Example :- Prove that $A \cup (B \cup C) = (A \cup B) \cup C$ by using set builder notation. Note X is a variable :-

Ans :- Take arbitrary element x in $A \cup (B \cup C)$:-

$x \in [A \cup (B \cup C)] \Rightarrow x \in A \vee x \in (B \cup C)$ definition of union (U)

$\Rightarrow x \in A \vee (x \in B \vee x \in C)$ definition of union (U)

$\Rightarrow (x \in A \vee x \in B) \vee x \in C$ associative (Logic)

$\Rightarrow x \in (A \cup B) \vee x \in C$ definition of union (U)

$\Rightarrow x \in [(A \cup A) \cup C]$ definition of union (U)

$\therefore A \cup (B \cup C) \Rightarrow (A \cup B) \cup C$ ✓✓

⑭ Example :- prove that $A - B = A \cap \overline{B}$ using set builder notation.

note X is a variable.

Ans :- take arbitrary element x in $A - B$:-

$x \in A - B \Rightarrow x \in A \wedge x \notin B$ definition of complement

$\Rightarrow x \in A \wedge \sim x \in B$ definition of $\notin$

$\Rightarrow x \in A \wedge x \in \overline{B}$ definition of complement

$\Rightarrow x \in A \cap \overline{B}$ definition of intersection (∩)

⑮ Example :- prove by set identities $(A \cap \overline{B}) \cup (A \cap B) = A \cap (B \cup \overline{B})$.

Ans :- $(A \cap \overline{B}) \cup (A \cap B)$

$= (A \cap \overline{B}) \cup (A \cap \overline{\overline{B}})$ by demorgan

$= (A \cap \overline{B}) \cup (A \cap B)$ by complement

$= A \cap (\overline{B} \cup B)$ by distribution

$= A \cap (B \cup \overline{B})$ by commutative

⑯ Homework :- Prove $A - (A - B) = A \cap B$ by using :- 1. Set identities, and

2. Set builder notations.